Eduardo Vasconcelos

Security Engineer | Adjunct Professor | PhD student (USP)

Linux kernel, Android security, software engineering, systems programming, R&D

Contact

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- **GitLab:** https://gitlab.com/vasconcedu

Key Highlights

- 10+ years of experience in software engineering, with emphasis on security engineering;
- Contributor to various open source projects, focusing on the Linux kernel;
- Adjunct professor with experience mentoring and delivering higher education courses to CS undergraduate students;
- MSc in Computer Science (USP) recipient of the Distinguished Paper Award at CBSOFT'24;
- Computer Engineer (USP) recipient of the school's Best Computer Engineering Undergraduate Thesis Award 2016;
- Proficient in three languages: Portuguese (*native*), English (*IELTS: C1*), and French (*TCF: B2--C1*).

Skills

- **Advanced:** security engineering, software development, threat modeling, security testing, security code review, R&D
- **Intermediate:** reverse engineering, DevSecOps, cryptography, Linux systems administration, Linux kernel development, Machine Learning
- **Familiar:** incident response, embedded systems

Work History

- **Master Software Security Analyst** @ SiDi, Brazil (remote)

  Aug 2024--Present

  - Delivered threat modeling and security testing of Android applications and device management system software to a global Android OEM client;
  - Conducted research aimed at proposing novel application hardening approaches for Android memory corruption vulnerabilities in third-party native libraries;
  - Reverse engineered Android malware using JADX and Ghidra, decompiling DEX to Java and analyzing x86 shared objects to develop malicious behavior heuristics for malware detection;
  - Designed and implemented ML pipelines for Android malware classification in Python;

  - Designed and implemented automated Android malware analysis tooling in Python to unveil and monitor malicious application behavior;
  - Participated in client AI red teaming task forces, contributing to AI security testing knowledge building and reporting security vulnerabilities in AI agents;
  - Delivered AI security training to 30+ resident tech students, including AI security engineering and red teaming;
  - Proponent and key contributor in revitalizing the organization's Security Champions program.

- **Adjunct Professor** @ UNIP, Brazil (on-site)

  Aug 2026--Present

  - Adjunct Professor of Computer Science;
  - Teaching OOP to 30+ undergraduate students using Java;
  - Teaching Integrated Project Development (Embedded Systems), mentoring 30+ undergraduate students building IoT prototypes using ESP32, Python and SQLite.

- **Open Source Contributor** @ various projects

  Apr 2021--Present

  - Contributed to: the Linux kernel, AppArmor, the OWASP AI Testing Guide, and the OWASP MASTG;
  - Currently focusing on the Linux Security Module (LSM) kernel framework.

- **Technical Instructor** @ SENAI, Brazil (on-site)

  May 2025--Jan 2026

  - Taught programming to 15+ vocational students per class using Python, guiding students through practical programming projects;
  - Taught business intelligence development using Microsoft Power BI, leading students in building interactive dashboards from datasets.

- **Senior Mobile Security Engineer** @ iFood, Brazil (remote)

  Feb 2022--Aug 2024

  - Designed and implemented security and fraud prevention mechanisms in C++, integrated into mission-critical Android applications via JNI, serving 40M+ monthly users;
  - Reverse engineered malicious Android applications using JADX and Ghidra, decompiling DEX to Java and analyzing x86 shared objects to unveil methods and behaviors used by fraudsters in food delivery services;
  - Integrated RASP solutions into GitLab CI/CD pipelines for mobile applications.

- **Senior Software Security Analyst** @ SiDi, Brazil (remote)

  Apr 2019--Feb 2022

  - Delivered ongoing offensive security testing services to a global Android OEM client, emphasizing Android applications and system services;

    - Conducted research on Android API and system vulnerabilities, including issues related to Android accessibility services abuses and permission circumvention.

- **Information Security Analyst** @ Hacker Rangers, Brazil (on-site)

  Jun 2017--Apr 2019

    - Delivered web application and network security assessments and phishing campaigns for private sector clients;
    - Developed the Hacker Rangers SaaS security awareness training platform;
    - Administered IBM Security Guardium database audit and protection virtual appliances for private sector and government clients, covering deployment, configuration, and operations.

- **Information Technology Intern** @ Embraer, Brazil (on-site)

  Sep 2016--Jun 2017

    - Contributed to the development of an IT change management plan;
    - Developed tooling to analyze corporate web traffic in Python.

- **Engineering Intern** @ SigMEDIA, Ireland (on-site)

  May 2014--Jul 2014

    - Developed an ML ensemble model to detect laughter in speech;
    - Developed tooling to process digital voice recording datasets in Python.

Education

- **Doctoral Degree in Computer Science** @ USP, Brazil

  Apr 2026--Present

    - Research topic: evaluating the energy cost of security mechanisms in the Linux kernel.

- **Master's Degree in Computer Science** @ USP, Brazil

  Feb 2022--Aug 2025

    - Grade: A;
    - Research topic: leveraging Mutation Testing to evaluate Android security testing procedures;
    - Recipient of the Distinguished Paper Award at CBSOFT'24;
    - Served as a teaching assistant for Software Engineering.

- **Postgraduate Degree in Software Engineering** @ Unicamp, Brazil

  Feb 2019--Dec 2019

    - Grade: 9.4;
    - One-year Postgraduate Degree program in Software Engineering.

- **Engineer's Degree in Computer Engineering** @ USP, Brazil

  Feb 2012--Jul 2017

  - Grade: 7.5;
  - Recipient of the São Paulo state government's FAPESP undergraduate research scholarship;
  - Recipient of the school's Best Computer Engineering Undergraduate Thesis Award 2016;
  - Served as a teaching assistant for Python programming;
  - Served as a teaching assistant for Java and C++ OOP;
  - Served as a volunteer PHP developer at the school's junior enterprise.

- **Visiting Student in Electronic Engineering** @ TCD, Ireland

  Sep 2013--May 2014

  - Full year visiting student;
  - Achieved first class honors;
  - Recipient of the Brazilian federal government's Science without Borders international student exchange scholarship;
  - Winner at the Trinity Engineering Student Entrepreneurs of the Year Competition 2014.

- **Professional Training in Computer Hardware and Network Maintenance** @ Microlins, Brazil

  Mar 2007--Sep 2007

  - Six-month professional training program in computer hardware, operating systems, and computer networks.